

Effect of diets and light regimes on calcification and somatic growth of the sea urchin *Tripneustes gratilla elatensis*

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ABSTRACT

The effects of *Ulva lactuca* (cultivated in the outlet of an Integrated Multi-Trophic Aquaculture [IMTA] system) vs. formulated diets on the somatic growth of the sea urchin *Tripneustes gratilla elatensis* were evaluated under three light regimes (continuous light, continuous darkness and ambient dark-light cycle). Total CaCO_3 precipitation and calcification rate, test diameter and surface area were used as growth parameters. The different light regimes tested did not appear to have a significant effect on any of these parameters. Sea urchins exposed to ambient light had significantly faster growth (measured as wet weight) than those exposed to continuous darkness or continuous light, a phenomenon attributed to gonad growth at the expense of shell growth. IMTA-produced *U. lactuca* was the preferential food source for promoting calcification, as average calcification rates in sea urchins fed formulated and *U. lactuca* diets, $(1.31 \pm 0.69$ and $2.1 \pm 0.31 \text{ mg h}^{-1} \text{SU}^{-1}$, respectively), were significantly higher in the urchins fed *U. lactuca*. Calcification rates in adult *T. gratilla elatensis*, measured as percentage of skeleton added per day, decreased with the juveniles' increase in size, as larger sea urchins allocated most of the CaCO_3 in favour of skeleton density rather than somatic growth. A 3D laser scanner was used to measure test surface area, from which significant correlation could be extrapolated extremely accurately between test diameter and the sea urchin surface area, volume, and skeletal and test dry weight, while alkalinity depletion in the surrounding sea water was used to calculate total calcium carbonate precipitation and calcification rates in the sea urchin. Both measurements applied for the first time to sea urchins, can be used to monitor the progress of growth in an industrial echinoculture setting.

1. Introduction

Cost-effective feeds that maximise somatic and gonadal growth while promoting commercially desirable gonad colouration and taste are critical for the establishment of a viable sea urchin aquaculture industry (Shpigel et al., 2005). Disadvantages of using macroalgae collected from the wild as feed often include limited availability, restricted harvesting periods, and temporal and seasonal variation in quality (Shpigel et al., 2018). Apart from the negative environmental impact that the removal of seaweeds may produce, the relatively low protein content of the wild-harvested macroalgae does not support maximal somatic and gonadal development of sea urchins (Troell et al., 2006; Anderson et al., 2007; Cook and Kelly, 2007). Use of macroalgae from the wild alone is therefore unlikely to be economically sustainable for large-scale commercial echinoculture. Conversely, formulated feeds can be enriched with essential ingredients and promote higher sea urchin production (Lawrence et al., 1992; Troell et al. 2006). When properly formulated, protein rich feeds can enhance gonad quality of various sea urchin species (Fernandez and Boudouresque, 1998; Robinson et al., 2002; Pearce et al., 2004; Shpigel et al., 2005; Hammer et al., 2006; Cyrus et al., 2015; Prato et al., 2018). However, production of formulated diets is relatively expensive and should be tailored to the

specific requirements of each sea urchin species. To the best of our knowledge, no commercial cost-effective feeds formulated for *Tripneustes* spp. are available today on the market.

Cultured *U. lactuca* is a fast growing macrophyte used as a fresh food source in sea urchin culture (Shpigel et al., 2005; Cyrus et al., 2014; Shpigel et al., 2018). Either fresh or incorporated in dry pellets, it can lead to the development of high quality gonads in *T. gratilla* (Cyrus et al., 2015) and *T. gratilla elatensis*, a tropical Indo-Pacific subspecies of *T. gratilla* (Shpigel et al., 2018). *T. gratilla* in countries such as the Philippines, Fiji (Andrew and Agatsuma, 2002), Australia (Mos et al., 2012), and Japan (Rahman et al., 2009) is regarded as one of the most promising candidates for aquaculture (Rahman et al., 2009; Mos et al., 2012; Shpigel et al., 2018). *T. gratilla elatensis* holds a great commercial potential due to its fast growth rate and high roe quality (Lawrence and Bazhin, 1998). *U. lactuca* is cultivated in Eilat (Israel) as part of an Integrated Multi-Trophic Aquaculture (IMTA) system and functions as a biofilter for the purification of the nutrient-rich effluents from fishponds (Ben-Ari et al., 2014). *U. lactuca* produced in IMTA has shown a relatively fast growth rate, presumably due to the continuous steady supply of nutrients from the fishponds and minimal disturbance by grazers and epiphytes (Mata et al., 2010; Abreu et al., 2011; Silva et al., 2015). It contains high protein levels (up to 40% dry weight - DW) and

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high stable lipid content throughout the year (Shpigel et al., 2018).

An additional aspect that has been studied extensively concerns the effect of light regimes on gonad development, sexual maturity and suspension of gametogenesis. Different light regimes were used as effective tools for manipulating the gonad cycle (Pearse et al., 1986; McClintock and Stephen, 1990; Spirlet et al., 2000; Shpigel et al., 2004; Siikavuopio, 2014). However, the effect of photoperiod and food quality on test calcification rate, which would be important for an echinoculture industry, has received lesser attention.

Most of the studies that evaluated the sea urchin's somatic growth used test diameter and wet weight (WW). However, WW is not a precise parameter for somatic growth evaluation since it includes gonad weight, which is extremely variable throughout the year and can change by up to 30% (Dincer and Cakli, 2007; Muthiga, 2005). As calcifiers, sea urchins rely on calcium and carbonate ion availability for the construction of their calcium carbonate skeleton (Dafni and Erez, 1987). Calcium has been shown to play an important role in the regeneration process of sea urchins' spines and teeth (Heatfield, 1971). The effects of the anthropogenic elevation of atmospheric carbon dioxide and the resulting ocean acidification together with global warming on marine organisms, particularly on their calcification mechanisms, have been getting increasing attention (Ries et al., 2009; Sheppard et al., 2010; Yeruham et al., 2015). In contrast to the relatively adequate knowledge on sea urchin calcification at the larval stages, quantitative studies on calcification rates in adult sea urchins appear to be few. The technique of alkalinity depletion was first used to evaluate the effects of water acidification on CaCO_3 calcification rates in corals, using the Total Alkalinity (TA) method (e.g. Smith, 1973). Since calcification rate directly affects the somatic growth of sea urchins, we hypothesised that measuring alkalinity depletion in the organism's immediate surroundings could be a useful additional tool to assess the sea urchins' response to various feed and environmental conditions.

In the present study, the effects of light regimes and feed (IMTA-produced *U. lactuca* and formulated diets) on *T. gratilla elatensis*' somatic growth were evaluated using total CaCO_3 precipitation, calcification rate, Relative Test Density (RTD), test diameter and surface area as parameters.

2. Material and methods

The sea urchins used in the experiments originated from the same cohort of a second-generation *T. gratilla elatensis* broodstock that had been collected from the Gulf of Eilat (Red Sea, Israel). The sea urchins were acclimated to the experimental conditions for two weeks prior to the start of the experiments.

Two sets of experiments were carried out:

- 1) The effect of light regimes on calcification rate and somatic growth (96-day monitoring, from November 12, 2015 to February 16, 2016).
- 2) The effect of formulated diet vs. IMTA produced *U. lactuca* diet on somatic growth and calcification rate, RTD (75-day monitoring, from March 2 to May 15, 2015).

2.1. The effect of light regimes on calcification rate

Three light regimes were tested:

- 1) 24-h continuous light ('Light');
- 2) 24-h continuous darkness ('Dark');
- 3) Ambient dark-light cycle as control ('Ambient light');

For each treatment, six sea urchins (as replicates) (55 ± 5 mm test diameter, 64 ± 9 g wet weight) were used. Each sea urchin was randomly selected and placed in a 2-l transparent plastic jar with its own

water and air supply. For treatments 1 and 2, the sea urchins jars were held in 2 black-walled 40-l plastic tanks covered with a lid. In treatment 1, the tank was illuminated with a controlled daylight fluorescent lamp (9 W; 50 cm long). Light intensity was kept at 800–1000 lx at the surface level. The system was set up (and verified) in a way that the light was distributed evenly to the jars in all the tank area. For the ambient light treatment (3), the sea urchins were each placed in a 2-l transparent plastic jar inside white-walled 40-l plastic tanks without a cover (12–8 h and 11–5 h light in November and February respectively). Water flow rate for each jar was kept at 10 l h^{-1} in all treatments. Temperature and oxygen levels, measured twice a week at 08:00 and 15:00 in all the tanks throughout the entire experimental period, were maintained at $22 \pm 2^\circ\text{C}$ (SD) and $90 \pm 2\%$ (SD) oxygen saturation, respectively. The sea urchins were fed daily ad libitum with fresh cultured IMTA-produced *U. lactuca* as described in Shpigel et al. (2018). Faeces and uneaten food were removed daily from the jars by siphoning.

All the sea urchins' wet weight (WW), test diameter (TD), specific growth rate (SGR) and calcification rates (as described in Section 2.3) were measured every 20 days over 96 days. Surface area (SA) was calculated using the allometric relationship between TD to SA (as described in Section 2.6.5). Sea urchins' skeleton, test, spines and Aristotle's Lantern were measured at the end of the experiment.

2.2. The effect of formulated diet vs. seaweed diet on calcification rate

Two types of food were compared: fresh IMTA-produced *U. lactuca* and a formulated diet containing 10% dry IMTA-produced *U. lactuca* as described by Shpigel et al., (2018). Six sea urchins (test diameter size 66 ± 3.5 mm, wet weight 110 ± 17 g) in three replicates (three groups of six per replicate) were used in each treatment, as per the experimental setting described in paragraph 2.1. Water flow rate for each jar was kept at 10 l h^{-1} in all treatments. Temperature and oxygen levels, measured twice a week at 08:00 and 15:00. Temperatures throughout the entire experiment period were $24 \pm 2^\circ\text{C}$ and oxygen concentration was $90 \pm 2\%$.

The sea urchins were fed daily ad libitum with fresh cultured IMTA-produced *U. lactuca* and formulated diets as described in Shpigel et al. (2018). Faeces and uneaten food were removed from the jars daily.

Sea urchins' WW, TD, SGR were measured every 15 days over 75 days. CR was measured (as described in Section 2.3), every 15 days over 60 days. For each measurement period, six sea urchins, two per replicate, were randomly selected. SA was calculated using the allometric relationship between TD to SA (as described in Section 2.6.5). At the end of the experiment gonad weight (GW), skeleton, test, spines and Aristotle's Lantern were measured.

The formulated diet was based on the diet developed at the National Center for Mariculture (NCM), Eilat, Israel, that produced the most satisfactory results with *T. gratilla elatensis* (Shpigel et al., 2018). The total protein and lipid levels of our diet were similar to those measured in the formulated diets developed for *T. gratilla* by Cyrus et al. (2014), but with the ratio of IMTA-produced *U. lactuca* being 10% (instead of 20%). The ingredients of the formulated diet were mixed for 1 h in a 25-L batch mixer and 10-mm diameter pellets were produced using a laboratory-model California Pellet Mill (Crawfordsville, IN) with a steam pre-treatment unit. The moist pellets were sun-dried and stored at -20°C until utilisation (Table 1).

2.3. Calcification rate measurements

Calcium carbonate depletion rate was calculated according to the equation below and presented as $\mu\text{mole CaCO}_3 \text{ cm}^{-2} \text{ h}^{-1}$ (Schneider and Erez, 2006).

$$\frac{\Delta \text{Al}}{2} (V_{\text{Chamber}} - V_{\text{sea-urchin}}) \cdot 1000 \cdot 1.028$$

$$T \cdot SA$$

Table 1

(a) Ingredient composition (gr per 100 g dry matter) of the formulated diet. (b) Proximate biochemical composition (in % dry weight) of the diets ($n = 3$). Ca and Mg were measured using polarized X-rays: TURBOQUANT.

a. Ingredients		gr/100 g dry matter
Rapeseed		20.9
Wheat, whole		20.9
Soy bean		20.9
Corn starch		14.9
IMTA-produced <i>Ulva</i>		10 (37% protein)
Alginate		4
Linseed oil		2.9
Lecithin		2
Vitamin & mineral mix		2
Calcium carbonate		1.5
b. Biochemical composition		
	Formulated diet (10% <i>Ulva</i>)	IMTA- produced <i>Ulva</i>
Protein (%)	28.2 ± 1.0	33.4 ± 0.9
Lipid (%)	2.68 ± 0.07	1.16 ± 0.1
Ash (%)	12.6 ± 2.1	18.5 ± 2.6
Carbohydrate (%)	54.7 ± 12.5	47.4 ± 11.7
Ca	1.5%	2.1 ± 0.003
Mg	0.002%	6.1 ± 0.02

where ΔA_T represents the delta in alkalinity before and after incubation, V is the volume in liters, 1.028 g ml^{-1} is the density of seawater in the northern Gulf of Eilat, T is the time in hours and SA is the surface area of the skeleton in cm^2 . A_T was measured in three replicates of t_0 and t_5 for each individual by weighing 15 g seawater from each sample. Samples were titrated with 0.025 N HCl using Mettler Toledo DL15 titrator and applying Gran's titration method (Gran, 1950, 1952). Water flow in all the individual jars was stopped for five hours, while aeration for each jar was continued to maintain dissolved oxygen saturation levels above 75%. Two water samples were taken at the beginning and two at the end (t_0 and t_5) of this time period to measure alkalinity (A_T) levels. Twenty-four hours after sampling, the sea urchins were weighed and measured for test diameter and volume. Volume was measured by weighing the amount of water displaced by the sea urchin divided by 1.028 (water density, g ml^{-1}).

2.4. Skeleton surface area measurement

To evaluate test surface area, a 3D laser scanner (NextEngine™) was used, as described in Raz-Bahat et al. (2009) and Sholts et al. (2010). At the end of both experiments, after measuring the sea urchin WW, TD and volume, the skeleton was cleaned using sodium hypochlorite (NaOCl) and air dried for 24 h. 28 sea urchins from both experiments

from the same sea urchin batch used in the calcification measurements were scanned using ScanStudio HD software (Figs. 1a, b).

2.5. Measuring protocol

2.5.1. Specific WW growth (SGR)

The specific WW growth ($\% \text{ d}^{-1}$) rate (SGR), was calculated as:

$$\text{SGR} = 100 \times \ln (W_T/W_0)/t$$
 where W_0 = initial biomass, W_T = final biomass; t expresses the number of days of culture in the experimental set. Yield was calculated as the difference between initial and final weights and expressed in units of $\text{g m}^{-2} \text{ day}^{-1}$.

2.5.2. Skeleton DW measurement

The sea urchins were washed three times in distilled water, then submersed for 12 h in sodium hypochlorite (NaOCl), rinsed in distilled water and dried at 60°C to constant DW (~ 24 h).

2.5.3. Gonad somatic index (GSI)

GSI was calculated as:

$$\text{GSI} = (\text{gonads wet weight (g)} / \text{whole urchin wet weight (g)}) \times 100.$$

2.5.4. Relative Test Density (RTD)

RTD was measured as the ratio between dry test weight and the surface area (g cm^{-2})

$$\text{RTD} = \frac{\text{Test weight (g)}}{\text{Test surface area (cm}^2\text{)}}$$

as an estimate indication for the calcium carbonate deposited and relative skeletal density (Although it is known that density can vary with porosity of CaCO_3 deposition).

2.5.5. Allometric relationship

Skeleton, test, spines and Aristotle's Lantern weight and TD length parameters which were measured at the end of both experiments were used for the allometric relationship calculation between them. The outcome of these relationship equations was used to estimate the sea urchins' somatic performance throughout the experiments.

2.6. Statistical analyses

Figures are presented as mean values $\pm$ SD. Mean values of calcification rate, surface area, test diameter, SGR and WW were compared throughout the experiment. Gonad weight, GSI, RST were compared at the end of the experiment. For experiment 1, data were analysed using one-way analysis of variance (ANOVA) using the software programme, Prism-6 (GraphPad, USA). For experiment 2, data were analysed using t-test analysis. Differences were regarded as significant at $P < .05$.

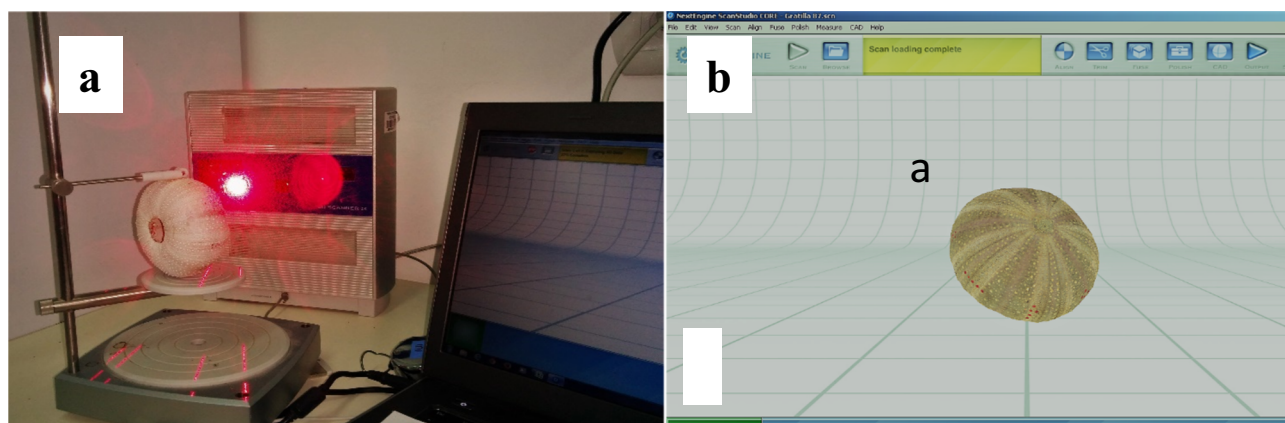


Fig. 1. NextEngine™ 3D scanner during scanning stage, *T. gratilla elatensis* skeleton placed on a rotating scanning stool in front of the scanner (a). A screen shot of the analysing software shows a 3D scanned skeleton (b).

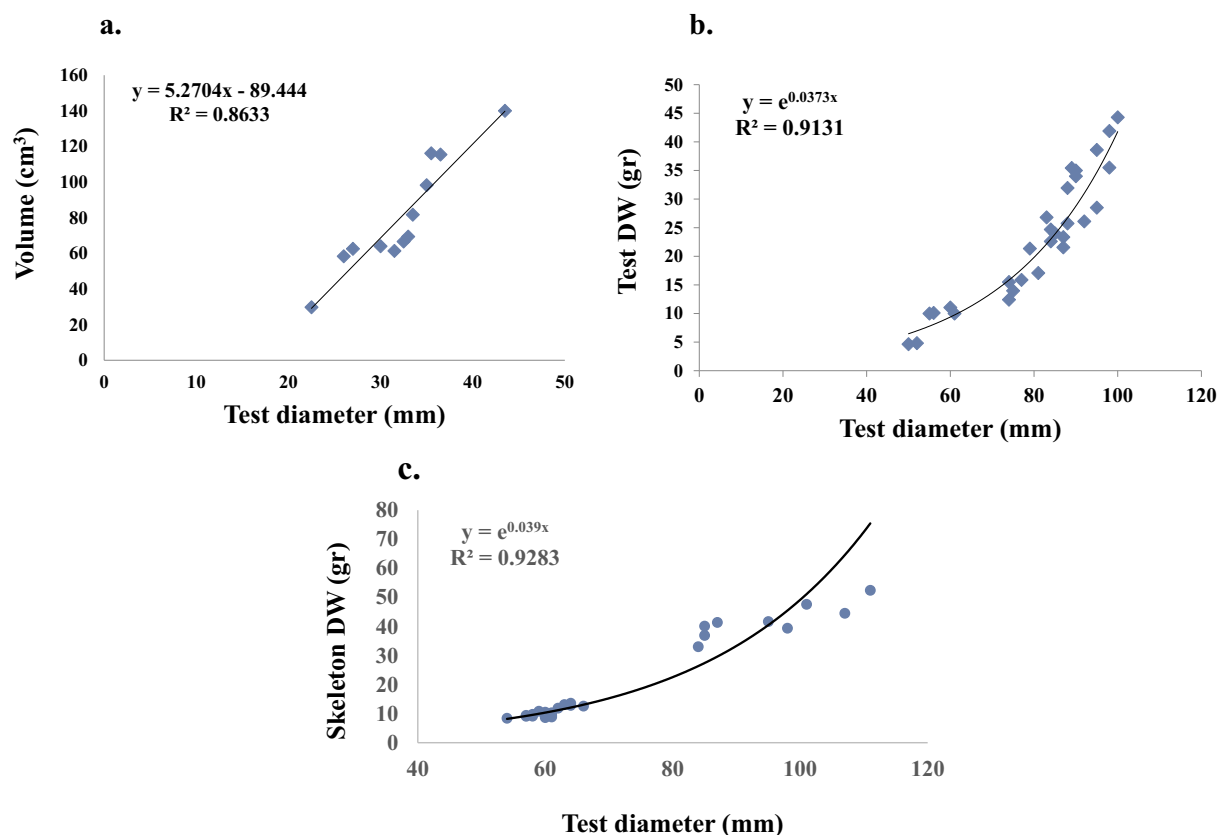


Fig. 2. Correlations among morphological parameters of *Tripneustes gratilla elatensis*. (A) Test diameter vs. volume ($n = 14$). (B) Test diameter vs. test DW, ($n = 28$). (C) Test diameter vs. skeleton DW ($n = 28$). The line equation and r^2 are shown in the figures.

Averages of the abiotic parameters, temperatures and oxygen levels were compared using non-parametric matched pairs. No transformations were performed on the data.

3. Results

3.1. Somatic parameters

Average sea urchin' dry weight (DW) to wet weight (WW) percentage was $12.4 \pm 1.4\%$; Sea urchin test weight to sea urchin WW was $9.8 \pm 1.1\%$. Test weight to sea urchin DW (excluding gonads) was $79.2 \pm 4.4\%$; Spine weight to sea urchin DW was $16.1 \pm 4.3\%$ and Aristotle's Lantern weight to DW ratio was $4.7 \pm 0.4\%$. Significant correlations ($P < .01$) were found between the sea urchins' test diameter and volume; test diameter and test DW (Figs. 2 a, b); test diameter and surface area; surface area and test DW; test surface to RTD (Figs. 3 a, b, c).

3.2. Light experiment

Survival rate was 100%.

3.2.1. Test diameter

Test diameter increased from 54.0 ± 4.94 , 54.80 ± 5.89 and 55.67 ± 5.05 to 59.50 ± 3.76 , 61.40 ± 3.36 and 61.5 ± 4.09 mm in the 'Dark', 'Light' and 'Ambient light' treatments, respectively. Average SGR (96 days) was 0.10 ± 0.07 , 0.11 ± 0.05 , $0.11 \pm 0.10\% d^{-1}$, in the 'Dark', 'Light' and 'Ambient light' treatments, respectively. No significant differences ($P > .05$) in test diameter growth and SGR were found among the treatments (Figs. 4 a, d, white bars).

3.2.2. Wet weight

The sea urchins' WW increased from 64.05 ± 9.42 , 63.30 ± 6.58 and 64.64 ± 4.51 to 77.32 ± 6.29 , 77.57 ± 6.15 and 90.80 ± 6.40 g in the 'Dark', 'Light' and 'Ambient light' treatments, respectively. Average SGR (96 days) was 0.19 ± 0.07 , 0.21 ± 0.07 , and $0.35 \pm 0.02\% d^{-1}$ in the 'Dark', 'Light' and 'Ambient light' treatments, respectively (Figs. 2 b, d, dark bars). On days 80 and 96, sea urchins exposed to the 'Ambient light' treatment grew significantly faster in WW (one-way ANOVA, $F = 4.006$; $P = .043$ and $F = 3.86$; $P = .044$, respectively) compared to those exposed to 'Dark' and 'Light' treatments.

3.2.3. Surface area

The sea urchins grew approximately 16.44 – 18.24 cm² in all the treatments, from an average of 65.66 ± 3.13 cm² to 83.48 ± 3.16 cm². No significant differences ($P > .05$) in surface area growth were found between treatments (Fig. 4 c).

3.2.4. Calcification rates

Average calcification rates throughout the 96-day experiment were 1.58 ± 0.22 , 1.43 ± 0.15 and 1.51 ± 0.16 mg CaCO₃ h⁻¹SU⁻¹ in the 'Dark', 'Light' and 'Ambient light' treatments, respectively. No significant differences ($P > .05$) in calcification rates were found among treatments (Fig. 5).

Total measured CaCO₃ precipitation in the sea urchins was 3.6 ± 0.12 , 3.29 ± 0.21 and 3.66 ± 0.46 g in the 'Dark', 'Light' and 'Ambient light' treatments, respectively (Table 2). Using the allometric relationship between radius to skeleton weight (from Figs. 2c), skeleton growth throughout the experiment was throughout the experiment was 2.87 , 2.45 and 2.24 g in the 'Dark', 'Light' and 'Ambient light' treatments, respectively (Table 2).

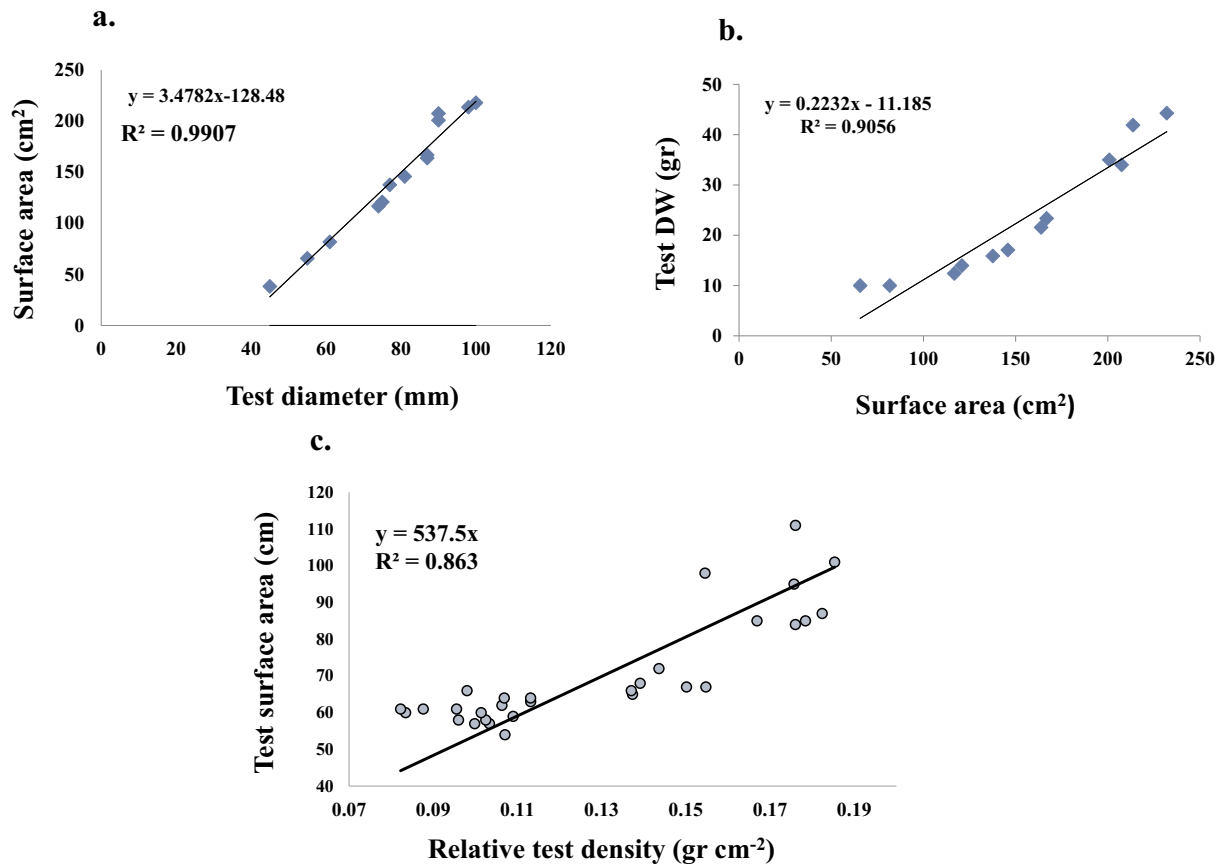


Fig. 3. Correlations among morphological parameters of *Tripneustes gratilla elatensis*. (A) Test diameter (mm) vs. scanned test surface area; $n = 14$; (B) Surface area vs. test DW; $n = 14$. (C) RTT vs. test surface area; $n = 28$. The line equation and r^2 are shown in the figures.

3.2.5. Relative Test Density (RTD)

RTD of the sea urchins were 0.09 ± 0.04 ; 0.09 ± 0.09 and 0.10 ± 0.03 g cm⁻² in the 'Dark', 'Light' and 'Ambient light' treatments, respectively. No significant differences ($P > .05$) in calcification rates and RTD were found among treatments (Fig. 5; Table 2).

3.3. Feeding Experiment

The sea urchins' survival rate was 93%. No different between the treatments.

3.4. Test diameter

Test diameter increased from 66.2 ± 4.07 and 67 ± 4.32 to 67.6 ± 1.88 and 70.5 ± 1.41 mm in sea urchins fed formulated and IMTA-produced *U. lactuca* diets, respectively (Fig. 6a). On days 60 and 75 (end of the experiment) test diameter was.

significantly higher in *U. lactuca* treatment than the formulated feed treatment (df 6, t -test, $p = .023$; 0.025 and 0.027 , respectively). SGR for the test diameter was 0.021 ± 0.009 and $0.058 \pm 0.01\%$ d⁻¹ when fed formulated and fresh *U. lactuca* diets, respectively, the latter being significantly higher (df 6, t -test $p = .001$) (Fig. 6c).

3.4.1. Wet weight

After 75 days, the sea urchins' WW increased from 105 ± 13.76 and 108 ± 13.16 g to 149 ± 11.16 g and 161 ± 10.50 when fed formulated diet and *U. lactuca*, respectively (Fig. 6b). SGR was 0.46 ± 0.04 and $0.53 \pm 0.06\%$ d⁻¹ when fed formulated and *U. lactuca* diet, respectively (Fig. 6c). A significant difference (df 6, t -test, $p = .049$) was found in SGR.

3.4.2. Surface area

On day 60, final surface areas were 103 ± 9.46 and 109 ± 4.65 cm² in sea urchins fed formulated and *U. lactuca* diet, respectively (Fig. 6d). Surface area of sea urchins fed *U. lactuca* diet was significantly higher (df 6, t -test $P = .07$).

3.4.3. Volume

The sea urchins' volume increased from 260.1 ± 13.6 and 265 ± 10.11 cm³ to 266 ± 11.16 g and 279 ± 10.1 cm³ when fed formulated diet and *U. lactuca*, respectively. At the end of the experiments, volume was significantly higher (df 6, t -test; $P = .045$) in individuals consuming the *U. lactuca* diet.

3.4.4. Gonad weight and GSI

No significant differences ($p > .05$) were found in final gonad weight of sea urchins fed formulated diet (30.48 ± 11.65 g) compared to *U. lactuca* diet (35.52 ± 12.91 g) (Fig. 7a). However, final GSI level in the formulated diet (15.73 ± 2.06) was significantly higher (df 6, t -test, $P = .026$) than in sea urchins fed fresh *U. lactuca* (10.69 ± 1.53) (Fig. 7b).

3.4.5. Calcification rate

Average calcification rates were 1.31 ± 0.69 and 2.1 ± 0.31 mg h⁻¹ SU⁻¹ in sea urchins fed formulated and *U. lactuca* diets, respectively. From days 15 to 60 calcification rates were higher in sea urchins fed *U. lactuca* diet, and significantly higher (df 6, t -test; $P = .045$) by the end of the experiment (Fig. 7c).

Total measured CaCO₃ precipitation during the 75-day experiment was 1.93 ± 0.30 and 2.96 ± 0.85 g for the sea urchins fed formulated and *U. lactuca* diets respectively (Table 2). Using the allometric relationship between radius to skeleton weight (from Figs. 2c), skeleton

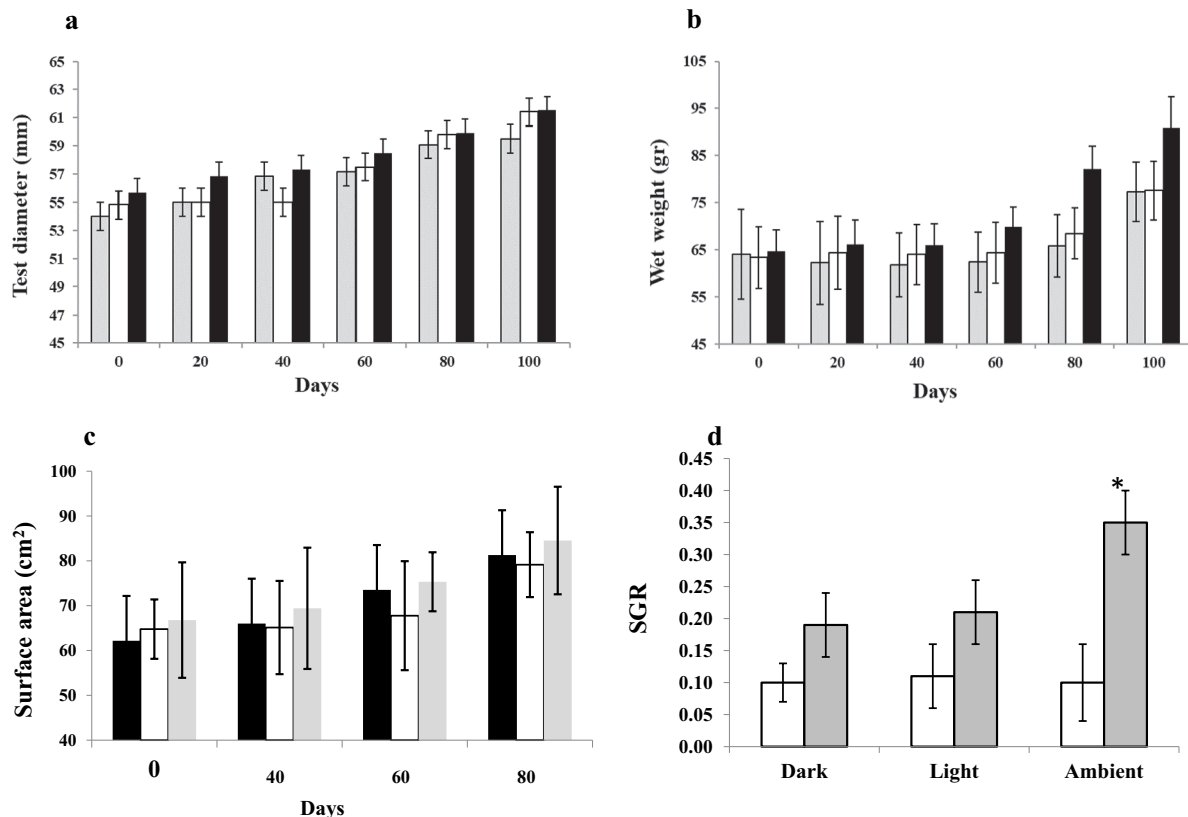


Fig. 4. Growth curves of *T. gratilla elatensis* exposed to continuous darkness (Dark), continuous light (Light) and ambient light conditions ($n = 18$). (a) Test diameter; (b) wet growth; and (c) surface area (using the allometric relationship). The black, white and grey bars represent dark, light and ambient treatments, respectively. (d) SGR of WW (grey bars) and test diameter (white bars). (Data are mean $\pm$ SD, $n = 18$). * represents significant differences ($P < .05$) between treatments.

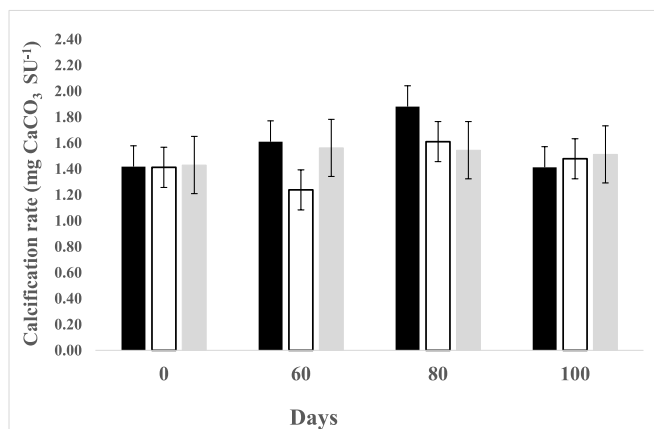


Fig. 5. Calcification rate of *T. gratilla elatensis* (by measuring calcium carbonate depletion) exposed to continuous darkness (Dark), continuous light (Light) and ambient light conditions presented in $\text{mg h}^{-1} \text{ SU}^{-1}$, at four sampling periods over 96 days. The grey, white and black bars represent continuous dark, continuous light and ambient treatments, respectively. (Data are mean $\pm$ SD, $n = 18$).

growth throughout the experiment was 0.79 and 1.89 g for formulated and *U. lactuca* diet treatments, respectively (Table 2).

3.4.6. Relative Test Density (RTD)

At the end of the experiment, RTD was 0.12 ± 0.02 and $0.15 \pm 0.03 \text{ g cm}^{-2}$ in the sea urchins fed formulated diets and *U. lactuca*, respectively (Fig. 7d; Table 2). RTD was significantly higher (df 6, t-test; $P = .015$) in individuals consuming the *U. lactuca* diet.

4. Discussion

Although our results showed a constant distribution of CaCO_3 weight in adult *T. gratilla elatensis* skeletal organs (79, 16 and 5% for test, spines and Aristotle's lantern, respectively), CaCO_3 precipitation (as biomass) was 30% (for ~55 mm sea urchins) to 45% (for ~66 mm sea urchins) higher than the actual biomass addition to skeleton growth. This was probably due to compensation for the constant erosion of the sea urchin's teeth. As shown by Dafni and Erez (1987), the Aristotle's lantern calcifies more rapidly than the other calcified organs. Reallocation of resources to different compartments (resulting in reduced skeleton dimensions and larger Aristotle's lanterns) occurs in conditions of low food availability, where resources are mobilised and preferentially allocated to the grazing apparatus (Lewis et al., 1990; Ebert, 1980). Apparently, total CaCO_3 precipitation and calcification rates among the skeletal organs do not have a constant distribution and depend on local conditions such as food availability, seasonality (Kobayashi and Tahi, 1969) and temperatures (Hermans et al., 2010). Calcification rates of *T. gratilla elatensis*, given as a percentage of skeleton added per day, decreased with the juvenile urchin's size (Dafni and Erez, 1987). Our results showed a similar pattern in adult individuals, with total CaCO_3 precipitation in the skeleton was higher and calcification rate being faster in sea urchins of ~60 mm in diameter than in those of ~70 mm in diameter (Table 2). However, these figures show that the process was ten times slower compared to the calcification rate of $2.75\% \text{ d}^{-1}$ obtained by Dafni and Erez (1987) for 11–13 mm diameter juveniles fed *Ulva* from the wild. RTD was 20–30% higher in sea urchins of ~70 mm in diameter than in those of ~60 mm (final lengths), suggesting that the larger sea urchins allocated most of the CaCO_3 to skeleton density rather than to somatic growth.

Table 2

Summary of *Tripneustes gratilla elatensis* performance *per individual sea urchin* exposed to different light regimes (Experiment 1) and diets (Experiment 2). Average wet weight (WW), test diameter (TD), CaCO_3 precipitation (CP), volume and surface area (SA) (measured). The relative test density (RTD) was calculated (according to 2.6.4). Skeleton growth was calculated using the allometric relationship presented in Figs. 2 and 3 (data are mean values $\pm$ SD; Δ is the mathematical difference between final and initial parameters).

Light Regime Exp. (96 days)	Initial WW (gr)	Δ WW (gr)	Initial TD (mm)	Δ TD (mm)	Initial volume (cm^3) Cal.	Δ Volume (cm^3) Cal.	Initial SA (cm^2)	Δ Test SA (cm^2)	Skeleton growth(gr)	Average CP (gr)	RTD (gr cm^{-2})
		Rate (gr d^{-1})		Rate (mm d^{-1})				Rate ($\text{cm}^2 \text{d}^{-1}$)		CP rate (mg d^{-1})	
Dark	64.1 $\pm$ 9.4	13.44 0.14	54.1 $\pm$ 4.9	5.47 0.057	195 $\pm$ 16	28.99	58.9 $\pm$ 7.2	19.13 0.19	2.87	3.64 $\pm$ 0.12 37.9	0.09 $\pm$ 0.04
Light	63.3 $\pm$ 6.6	14.4 0.15	54.8 $\pm$ 5.9	6.52 0.068	199 $\pm$ 21	28.72	61.7 $\pm$ 7.5	22.95 0.23	2.38	3.29 $\pm$ 0.21 34.2	0.09 $\pm$ 0.09
Ambient	64.6 $\pm$ 4.5	25.92 0.27	55.7 $\pm$ 5.1	5.85 0.061	203 $\pm$ 16	28.1	61.5 $\pm$ 9.6	20.29 0.21	2.24	3.66 $\pm$ 0.46 38.12	0.10 $\pm$ 0.03
Feeding Exp. (60 days)											
Form. diet	105.2 $\pm$ 14	28.8 0.48	66.2 $\pm$ 4.07	2.5 0.026	260 $\pm$ 13	6.1	102 $\pm$ 8.49	4.81 0.08	1.07	1.88 $\pm$ 0.48 31.33	0.12 $\pm$ 0.02
<i>U. lactuca</i> diet	108.1 $\pm$ 13	33.6 0.56	67.0 $\pm$ 3.84	3.1 0.031	265 $\pm$ 10	14	104 $\pm$ 6.41	7.28 0.12	1.69	3.02 $\pm$ 0.85 49.3	0.15 $\pm$ 0.03

4.1.1. Light experiment

The different light regimes tested did not appear to have a significant effect on any of the parameters for calcification (test diameter, surface area, calcification rate and RTD). Therefore, it seems safe to assume that the observation that sea urchins exposed to 'Ambient light' treatment grew significantly faster in WW than those exposed to 'Dark' and 'Light' treatments is related to gonad growth and not to shell growth. Light regime can have varying effects on gonad performance in

certain species (Pearse et al., 1986; Pearse and Cameron, 1991; Shpigiel et al., 2004). Under a long light regime, some species produced large gonads (Walker and Lesser, 1998), whereas in other species it was daylight shortening that promoted gonadal growth (Yamamoto et al., 1988). Previous studies showed no direct correlation between light and *T. gratilla* gonad development (Fouda and Hellal, 1990; Muthiga, 2005). However, in our study, the final WW of *T. gratilla elatensis* under ambient light condition was almost double compared to the 'dark' and 'light' treatments, suggesting that light effect on gonad development is species and/or site specific.

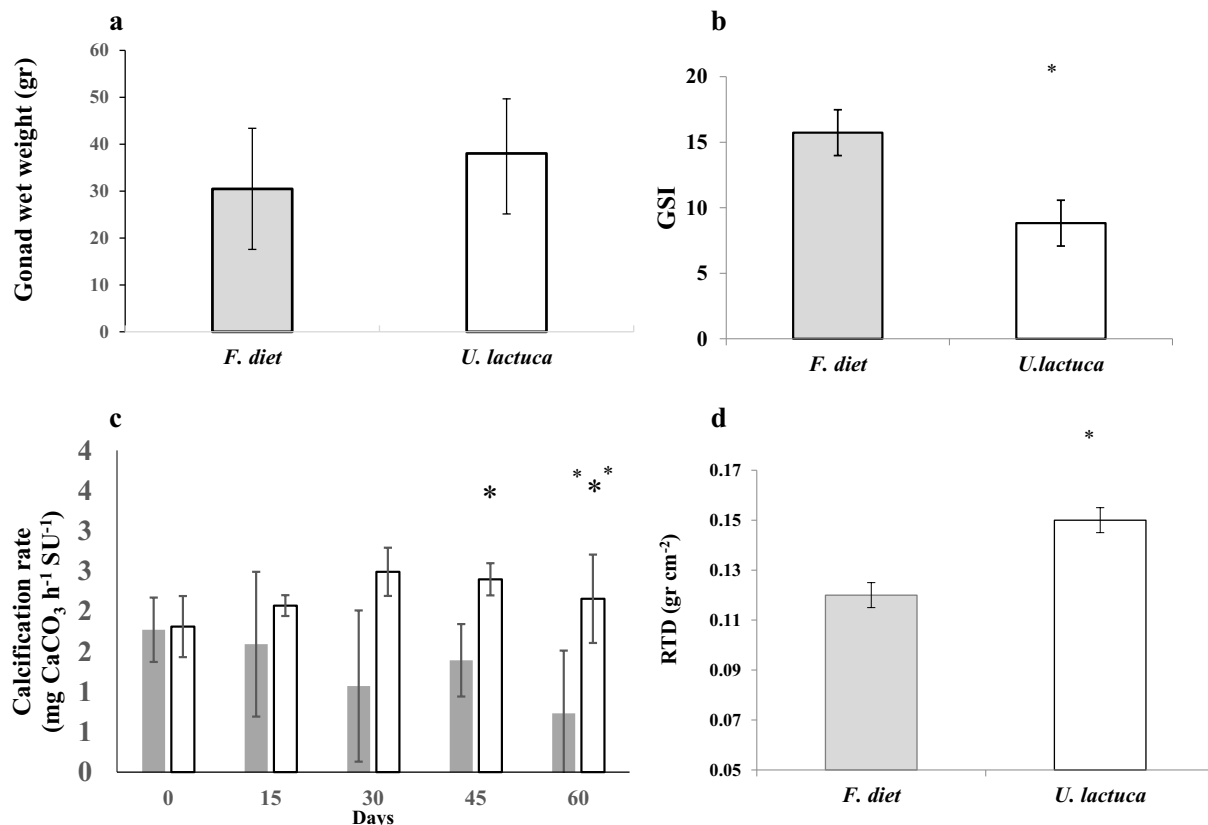


Fig. 6. Growth curves of *T. gratilla elatensis* exposed to *Ulva lactuca* (white bars) and formulated (grey bars) diets. (a) Test diameter, (b) wet weight, (c) SGR for diameter, and (d) surface area (using the allometric relationship). (Data are mean $\pm$ SD, n = 18). * represents significant differences ($P < .05$) between treatments.

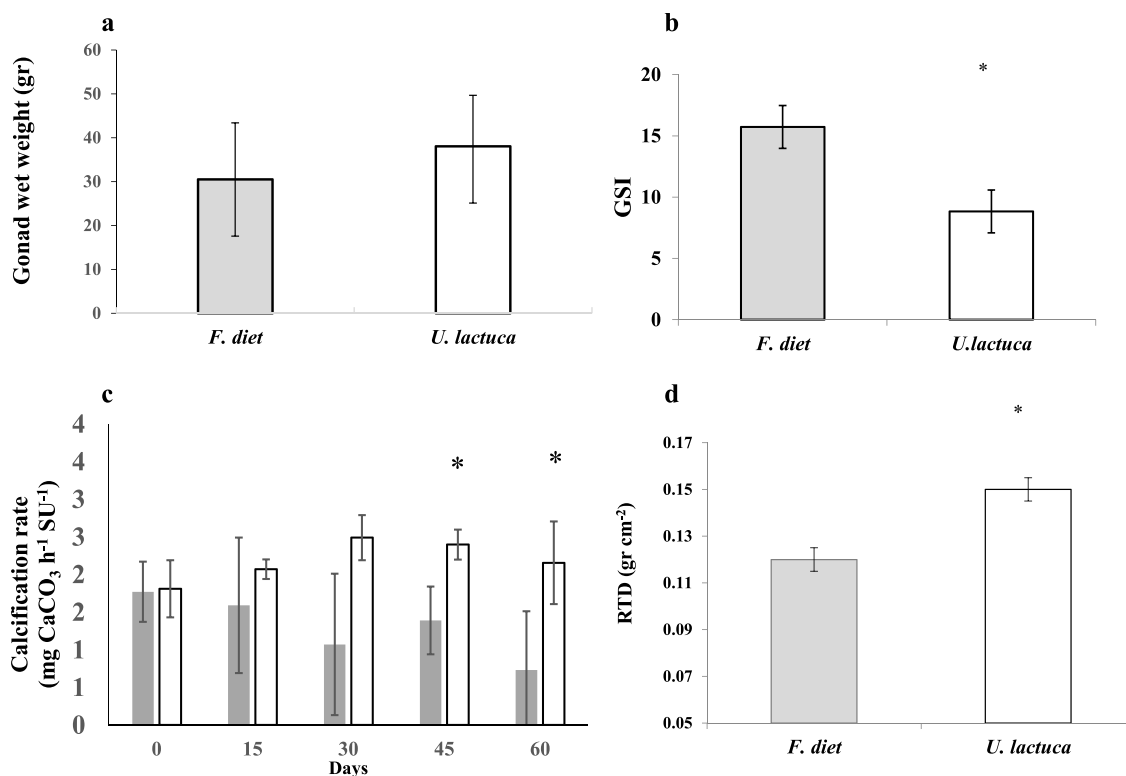


Fig. 7. Final gonad weight (a); GSI (b); calcification rate (c) by measuring calcium carbonate depletion; and RTD (d) of *T. gratilla elatensis* exposed to *Ulva lactuca* (white bars) and formulated (grey bars) diets. (Data are mean $\pm$ SD, n = 18). * represents significant differences ($P < .05$) between treatments.

4.1.2. Feeding experiment

Two growth patterns were observed: IMTA-produced *U. lactuca* supported both somatic (via efficient CaCO_3 precipitation) and gonad growth. Formulated diets, however, supported mainly gonad growth. Similarly, Cyrus et al. (2015) demonstrated that juvenile *T. gratilla* fed *Ulva armoricana* showed significantly better somatic growth compared with urchins fed formulated diets. GSI of sea urchins fed formulated diets was higher due to the sea urchins' slow somatic growth and the ratio between gonad weight and total weight.

Lawrence (2000) claims that somatic growth and gonadal growth in sea urchins are independent of each other: some nutrients support somatic and some others gonadal growth. For example, test and gonad growth in *Strongylocentrotus intermedius* fed different species of algae showed that *Ulva australis* promoted faster somatic growth than gonadal growth (Shuuluka, 2011). In the present study, IMTA-produced *U. lactuca* supported both growth patterns, suggesting that *U. lactuca* contains all the nutrients to support both somatic and gonadal growth. Similar results with *U. lactuca* were obtained by Shpigiel et al. (2018).

The presence of certain elements in the diets strongly influences the sea urchins' somatic growth (Kennedy et al., 2005). Calcium and magnesium, in particular, have been shown to be important elements in the production of the sea urchins' skeleton (Pearse and Pearse, 1975; Ebert, 2001; Asnaghi et al., 2014) and a lack of these elements in the formulated diets can lead to reduced somatic growth (Powell et al., 2010). It was also found that the uptake of calcium from the diet is fundamental in controlling sea urchin skeleton growth (Asnaghi et al., 2014). For example, wet weights and diameters of the sea urchins *Lyttechinus variegatus* fed ad libitum with 4.7% calcium in the formulated diet were significantly slightly higher than those fed 1.9 or 2.4% calcium feeds (Powell et al., 2010).

As various studies have shown, calcium and magnesium concentration in *Ulva* spp. is relatively high throughout the year (Shiber and Washburn, 1978; Ho, 1981; Ratana-arporn and Chirapart, 2006; Shuuluka, 2011; Szlag-Sikora et al., 2016). IMTA-produced *U. lactuca*

in the present study contained relatively high levels of Ca (2.5% of DW) and Mg (6% of DW) (Table 1). We can therefore assume that IMTA-produced *U. lactuca* is a good food source for somatic growth of *T. gratilla elatensis*.

Little is known about skeletal element incorporation pathways and skeletal element sources in echinoderms. Potential sources of calcium and magnesium are seawater.

and food. Lewis et al. (1990) showed that skeletal calcium of the echinoderm skeleton derives from seawater calcium. However, there is evidence that calcium and magnesium are also derived from the diet (Kennedy et al., 2005; Powell et al., 2010; Asnaghi et al., 2013; Asnaghi et al., 2014). Kaneko et al. (1981) reported absorption rates of dietary calcium as high as 30% of the calcium present in the feed in 30 mm *Hemicentrotus pulcherrimus*.

Since the sea urchins were fed ad libitum, the quality of the food, rather than amount consumed, was likely to have had an effect on sea urchin performance. However, it was found by Shpigiel et al. (2018) that assimilation efficiency of IMTA produced *U. lactuca* is superior to that of formulated diet by 15%. This difference does not affect gonad development and growth. In our experiment we used the same food.

With regard to the proximate biochemical composition of both natural and formulated diets, total carbohydrate, lipid and gross energy levels were much higher in the latter, while only protein level was higher in the IMTA-produced *U. lactuca* (33% vs. 28%). According to Cyrus et al. (2015), high protein (above 26%) feeds during the somatic growth phase of small adult *T. gratilla* are not necessary. It is unclear why *U. lactuca* increased somatic growth compared to our formulated diets. Possibly the formulated diet composition used, while high in certain nutrients beneficial for gonad growth, was too low in Ca and Mg levels to be suitable for better somatic growth. Kaneko et al. (1981) found that *Hemicentrotus pulcherrimus* absorbs calcium from seaweed powder and distributes it throughout the body via the coelomic fluid, but altogether, the mechanism by which sea urchins utilise calcium from *Ulva* spp. to build the skeletal body is still unclear.

4.1.3. 3 D. Laser scanning method

The 3D scanner technology was first used for studying coral structures (Raz-Bahat et al., 2009). To the best of our knowledge, this is the first time that the surface area of regular sea urchins was measured with this method, which produced much higher accuracy than that traditionally calculated on the basis of the animal's height and diameter or as oblate spheroid which is not constant in regular sea urchins (Ebert, 1988). Use of a 3D laser scanner allowed measurements of the test surface area (SA) of regular sea urchin *T. gratilla elatensis* with the highest degree of accuracy. The significant correlations observed between the test diameter and test surface area, and between volume and test DW, enabled rapid and accurate extrapolation of these parameters by measuring only the test diameter, and without having to sacrifice the animal. These correlations can therefore be used to analyse the somatic performance of this sea urchin species in an experimental or commercial culture set.

5. Alkalinity depletion

Measuring alkalinity depletion as an indicator of calcification rates in sea urchins provides a useful tool for monitoring skeleton growth in vivo. Reduced calcification rate can indicate unfavourable biotic or abiotic conditions. The technique of alkalinity depletion was first used to evaluate the effects of water acidification on CaCO_3 calcification rates in corals (Smith, 1973). The decrease in total alkalinity is caused by reduction in Ca and carbonate ions that are taken from the seawater for the formation of calcium carbonate in the calcified organs (test, spines and Aristotle's lantern). Experiments that were made using sea water labeled with ^{45}Ca support this assumption on *T. gratilla elatensis* (Dafni and Erez, 1987) and on *Strongylocentrotus purpuratus* (Lewis et al., 1990). To the best of our knowledge, this is the first time that this method was used to correlate it with CaCO_3 calcification rates in sea urchins.

In conclusion, it was found that for *T. gratilla elatensis*, IMTA-produced *U. lactuca* is the preferred food source for calcification rate and total calcium precipitation when compared to the manufactured feed used in the experiments. Dietary formulations better tailored to *T. gratilla* specific requirements are still needed to improve the somatic growth of this sea urchin. Ambient light regime is preferred for gonad growth. The direct relationship between the test diameters of regular sea urchins and skeletal organs allows monitoring of somatic growth without the need to sacrifice the animals. The results presented in this paper fill some important knowledge gaps for the industrial aquaculture of this species.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit author statement

Shpigel Muki: Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection. Development or design of methodology Writing- Reviewing and Editing, writing original draft

Jonatan Erez: Development or design of methodology; Writing- Reviewing and Editing.

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